

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – SHORT ANSWER TEST

Course Code:	CSA16	Course Title:	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL1, BL2)	Assessment:	Assessment Tool 1 – Short Answer Test
Weightage:	10% of CIA	Total Marks:	100 (5 Questions × 20 Marks)
CO1 Statement:	<i>Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2</i>		
Student Name:	SK. Rahamath ali	Reg. No.:	192365048
Section / Batch:		Date:	

Instructions to Students

- Answer all questions. Each question carries 20 Marks. Total: 100 Marks.
- Clearly show all requirements, process, operations.
- Use appropriate models, schema represented scenario wise
- Justify each step in different dimension specified and measures clearly.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Short Answer Test	Architecture of Data Warehouse, ETL, OLAP	Q1 – Q5	Q1 –20 Q2 –20 Q3 –20 Q4 –20 Q5 –20
GRAND TOTAL:			100 Marks

SECTION A – SHORT ANSWER TEST

1. Suppose that a data warehouse for City Hospital consists of the following four dimensions: patient, doctor, department, and time, and two measures visit_count and avg_charge. At the lowest conceptual level, avg_charge stores the actual consultation fee charged for a patient visit. At higher conceptual levels, it stores the average fee.

- Draw a snowflake schema diagram for the data warehouse.
- Starting with the base cuboid [patient, doctor, department, time], what specific OLAP operations should be performed to list the average consultation fee department-wise for each year?
- If each dimension has four levels (including all), how many cuboids will this cube contain (including base and apex cuboids)?

3. A smart city project is collecting data from multiple sources, including:

- Traffic sensors for vehicle counts and congestion levels.
- IoT devices for monitoring air quality, noise levels, and energy consumption.
- Public transportation systems for ridership data and punctuality.
- Citizen feedback from a mobile app.

Questions:

Fact and Dimension Design:

- Design a star schema to analyze traffic congestion patterns, energy consumption trends, and public transportation efficiency.
- What measures and dimensions would you include? Justify your design.

3. Suppose that a banking data warehouse contains two fact tables: Transaction_Fact and Loan_Fact, sharing dimensions customer, branch, time, and account_type.

- Transaction_Fact → transaction_count, amount
- Loan_Fact → loan_amount, interest

(a) Draw a galaxy schema diagram.

(b) Identify OLAP operations to obtain branch-wise average transaction amount and loan amount yearly.

(c) If each dimension has four levels, calculate total cuboids.

4. A manufacturing company monitors product quality and production efficiency. The data warehouse contains:

- Dimensions:
 - Product: Category, type, and batch.
 - Factory: Location, department, and line number.
 - Time: Year, quarter, month, and week.
 - Supplier: Region, rating, and contract type.
- Measures:
 - Defect rate.
 - Production units.
 - Supplier reliability percentage.

Questions:

- Roll-up: Summarize defect rates by product category for the current year across all factories.
- Drill-down: Focus on a specific factory to analyze defect rates by department and product type.
- Slice: Isolate production unit data for suppliers with a rating of 4+ across all regions.
- Dice: Analyze defect rates and supplier reliability for selected product categories and factory locations.

5. A retail chain manages inventory and sales data across multiple stores. The data warehouse contains:

- Dimensions:
 - Product: Category, subcategory, and SKU.
 - Store: Region, city, and store ID.
 - Time: Year, quarter, month, and day.
 - Customer: Membership tier, age group, and gender.
- Measures:
 - Inventory turnover.
 - Total sales revenue.
 - Units sold.

Questions:

1. Roll-up: Summarize inventory turnover by product category across all stores for the last quarter.
2. Drill-down: Focus on sales data for electronics in a specific region, analysing by subcategory and SKU.
3. Slice: Isolate sales revenue data for male customers in the Bronze membership tier across all stores.
4. Dice: Analyze sales revenue and units sold for selected cities and product categories during specific time periods.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	WR	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Concepts	25%	Demonstrates thorough accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation, lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Marks Evaluation:

Question No.	Understanding of Concepts (25%)	Explanation (25%)	Application (20%)	Organization (15%)	Accuracy (10%)	Clarity (5%)	Total Marks
Q1 (20 Marks)							
Q2 (20 Marks)							
Q3 (20 Marks)							
Q4 (20 Marks)							
Q5 (20 Marks)							
Overall Total (100)							

Faculty in-charge

Course Coordinator

Head of Department

Signature _____

Signature: _____

Signature: _____

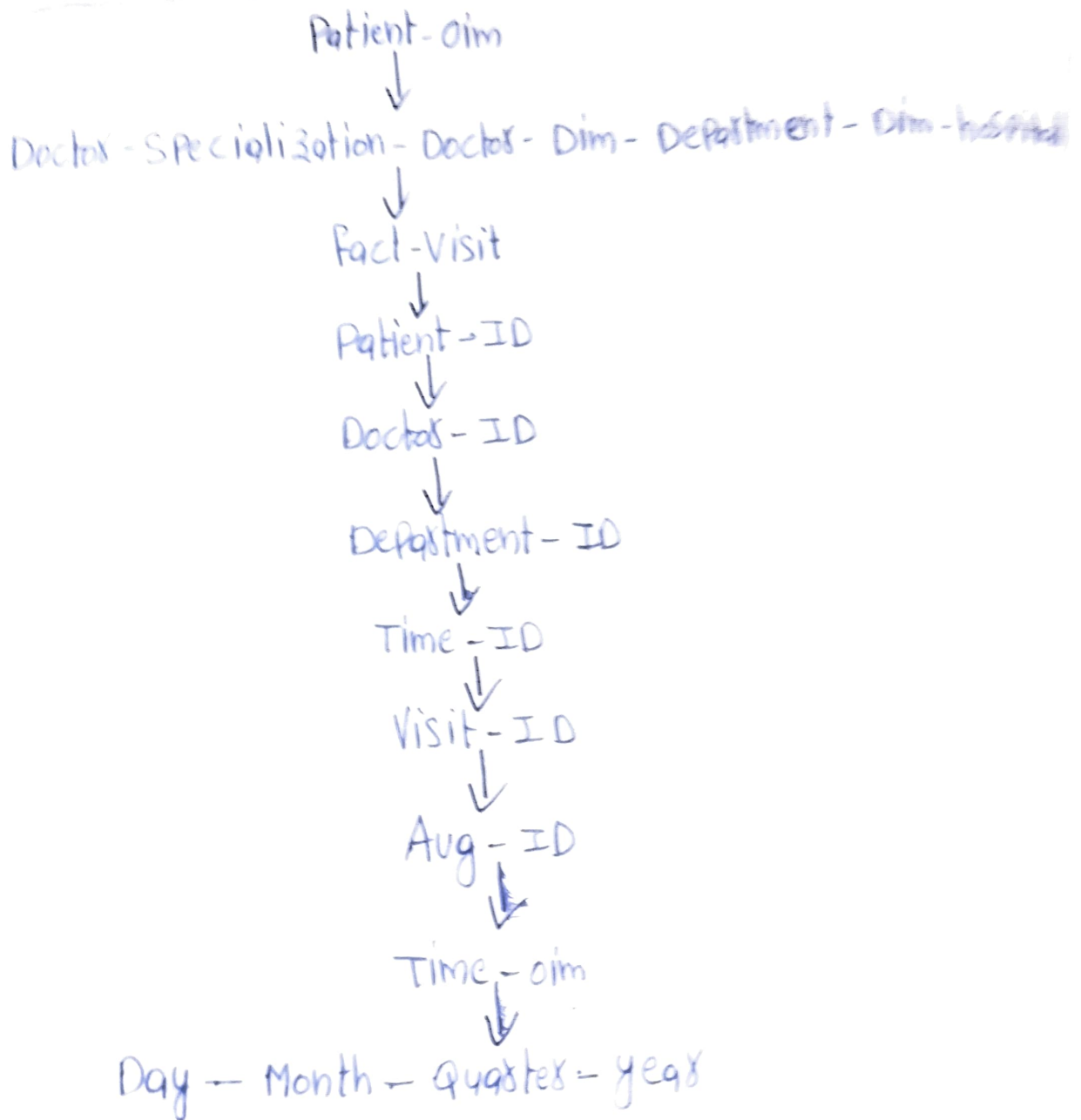
Date _____

Date: _____

Date: _____

City Hospital Data Warehouse

Starflake Schema



OLAP operations:-

- * Roll up Time $\rightarrow$ year
- * Roll up Patient $\rightarrow$ All
- * Roll up Doctor $\rightarrow$ All
- * Keep Department dimension unchanged.

* Aggregate Avg-charge

output:-

| Department | year | Average charge |

① Number of cuboids:-

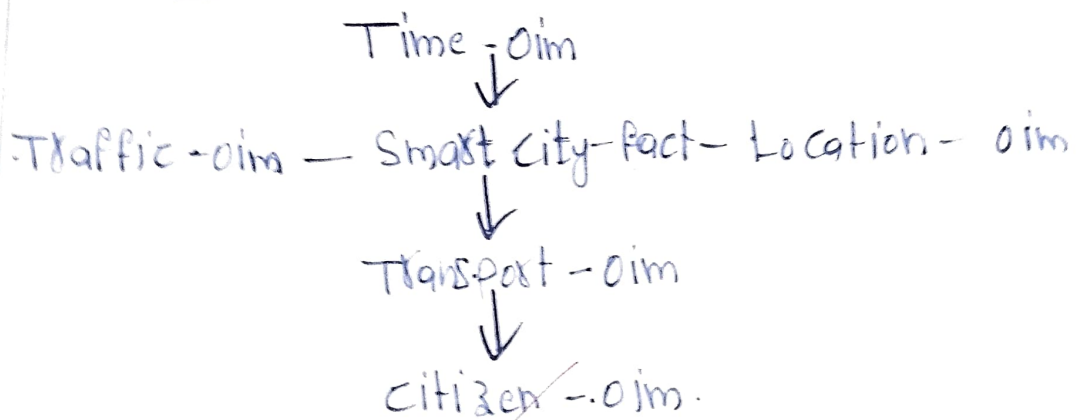
Formula:- Total cuboids = product of hierarchy levels.

Each dimensions = 4

$$4^4 = 256$$

② Smart city star schema:-

Star Schema:-



Fact table:-

<u>Measures</u>	<u>Time</u>	<u>Location</u>	<u>Traffic</u>	<u>Transport</u>
Vehicle count	Year	City	Sensor ID	Bus
congestion level	month	Zone	Road type	metro
Energy consumption	Day	area		

Air quality index hour

Noise level

Residence

Fractuality

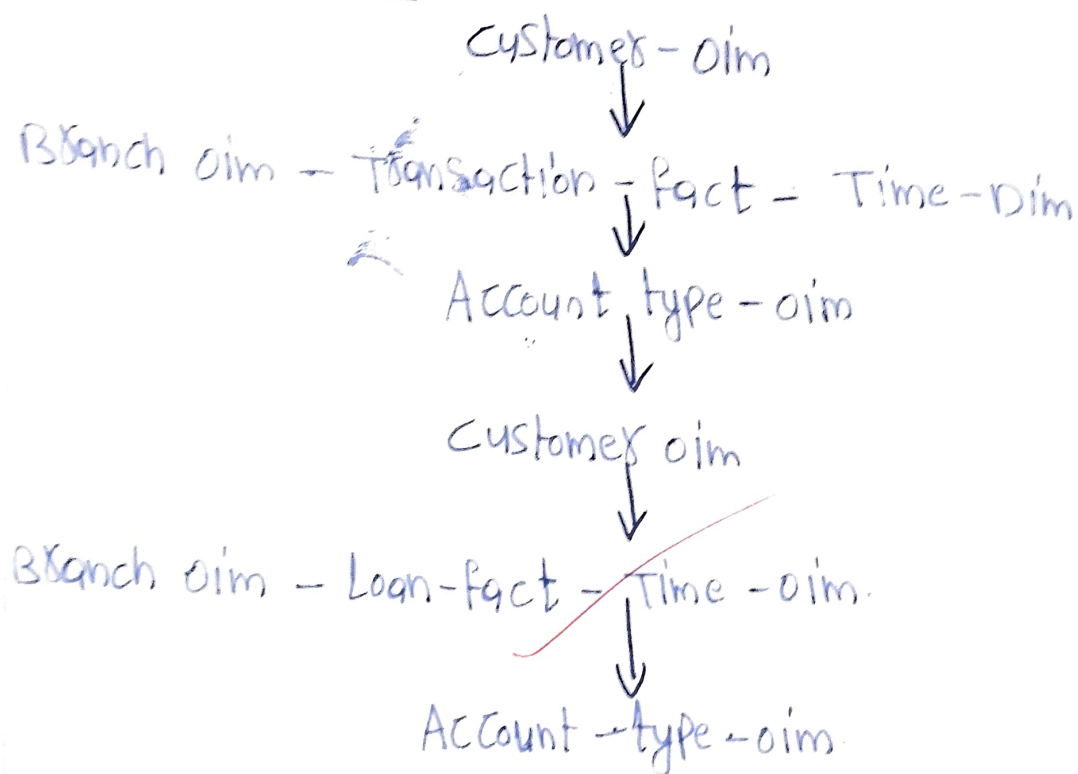
Justification:-

*measures store numerical values

*Dimensions enable multidimensional analysis by Location, time, transport and traffic.

③ Banking Data warehouse:-

④ Galaxy Schema:-



Transaction Fact

Transaction count
Amount

Loan Fact

Loan amount
Interest

⑥ OLAP operation:

Starling cyboid: Customer, Branch, time, Account type
operations:-

* Roll up Time $\rightarrow$ year

* Roll up Customer $\rightarrow$ All

* Roll up Account type $\rightarrow$ All

* Keep Branch

* Aggregate Average
Amount and loan
amount

Output:-

| Branch | Year | Avg transaction | Avg loan |

⑦ Cybolids

4 dimensions

each has 4 levels

$$4^4 = 256$$

④ Manufacturing Company :-

Roll-up

Roll-up Product :- Batch $\rightarrow$ type $\rightarrow$ Category

Roll-up Factory :- line $\rightarrow$ Department $\rightarrow$ Category

Keep current year.

Output :- Average defect rate by product category.

Drill-down Factory

Factory
↓
Department

Product
Category
↓
type

Slide :-

Condensation :- Supplier Rating ≥ 4

Display :- production unit.

Dice :-

Select

* Product categories

* Factory Location

measures

* Defect Rate

* Supplier reliability %

⑤ Retail chain:-

Roll-up:-

Roll-up:- SKU $\rightarrow$ Subcategory $\rightarrow$ category

Time:- Day $\rightarrow$ Month $\rightarrow$ Quarters

measure:- Inventory Turnover

Drill-down

Region



city



Store

Electronics



Sub-category



SKU

slice:-

Condition:-

Gender = male

membership = Bronze

measure:-

Sales Revenue

Dice:-

Selected

* cities

* Product categories

* Time Period

measure

* Sales Revenue

* units sold